

Industrial Security (Master)

Exercise Sheet – Section 05 Secure-by-Design Architectures

Exercise 5.1 – FEED-Phase Workshop

Imagine a greenfield 200-MW combined-cycle gas turbine. Draft a 1-page document for the FEED phase that:

- Lists the four zones with target SLs.
- Names every conduit + protocol + direction.
- Identifies one place where a data diode is appropriate and one where it is not.

Exercise 5.2 – Vendor SDL Checklist

Build the security clauses for a PLC vendor RFQ:

1. IEC 62443-4-1 conformance evidence.
2. SBOM format + delivery cadence.
3. Vulnerability handling SLA.
4. Long-term support window.
5. Signed firmware + signed program downloads.

Exercise 5.3 – Acceptance Test Procedure

Write the ATP for one of: (a) HMI workstation, (b) SCADA server, (c) site firewall. Include hardening checks, baseline traffic capture, vuln scan, backup/restore drill.

Exercise 5.4 – Brownfield Roadmap

For an existing 1990s refinery, produce a 5-year migration roadmap (Y1: visibility, Y2: segment + MFA, ...). Each year: deliverables, owner, cost band, risk reduced.